

Mohammadjavad Mehditabar

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SUMMARY

Machine Learning Engineer with 4 years of experience across academia and industry, specializing in applied R&D, data science, and the performance optimization of LLMs and Agentic AI. My work spans building end-to-end ML pipelines, benchmarking LLMs on cost-latency-accuracy trade-offs, and profiling hardware-level compute bottlenecks. I focus on aligning models and developing trade-off-aware agentic workflows to build highly efficient, scalable AI systems.

TECHNICAL SKILLS

Programming: Python, C#, Javascript, SQL

Frameworks & Tools: Git, Pytorch, Transformers, Pandas, Numpy, Docker, vLLM, LangChain

Research & Strategy: Qualitative Research, Survey Design, Customer Discovery, Business Modeling, AI Commercialization

EXPERIENCE

Graduate Research Assistant

Jan. 2025 – Present

SMART Lab, Dalhousie University (Advisor: Prof. Tushar Sharma)

Nova Scotia, Canada

- **LLMs Performance Benchmarking** [📄]:

- * Developed two novel, robust, and fair Multi-Criteria Decision Making (MCDM) methods with less than **5%** performance variation under severe noise conditions. Unlike static combination approaches, the methods dynamically learn trade-offs.
- * Evaluated **22** LLMs on performance and throughput trade-offs on the LiveCodeBench and CodeXGLUE benchmarks.
- * Profiled performance of quantized and standard language models across Hugging Face, Ollama, and vLLM, analyzing fundamental CUDA compute and memory bottlenecks during prefill and decode phases.
- * Investigated advanced serving techniques, including continuous batching, speculative decoding, and optimal batch sizing, using hardware insights to maximize GPU utilization for efficient multi-user deployment.

- **Code Inefficiencies Detection** [📄]:

- * Created a taxonomy of inefficiency patterns in code and curated a high-quality dataset of **3,000** efficient/inefficient code pairs, profiling time, energy, and memory consumption along with reasoning traces to enable downstream RL training.
- * Developed an agentic pipeline using a frontier LLM with dynamic context retrieval, achieving **94%** accuracy in identifying inefficiencies within arbitrary code snippets.

- **Meta-Data Aware Movie Recommender** [📄]:

- * Devised an end-to-end modified GPT-based recommender system fusing textual metadata with user interaction sequences via multi-task learning and LoRA fine-tuning.
- * Improved genre-awareness F1-score **37%** and boosted overall recommendation accuracy (NDCG) by up to **5%** using a custom genre-filtering algorithm and a decoupled ranking module.

R&D Analyst

Sep. 2025 – Dec. 2025

Lab2Market

Canada

- Conducted R&D and **80+** user interviews to evaluate technical solutions optimizing the cost-accuracy-latency trade-off in AI-based products.
- Validated **20** architectural hypotheses with stakeholders ranging from CXOs to developers, shaping novel value propositions for highly efficient and privacy-preserving Agentic AI systems handling sensitive data.
- Analyzed feedback to identify frequent operational bottlenecks and translated user challenges into actionable optimizations.

Research Assistant

Jul. 2021 – Jun. 2024

Iran University of Science Technology (Advisor: Prof. Sauleh Eetemadi)

Tehran, Iran

- **Consensus on Machine Learning Models** [📄]:

- * Improved the predictive accuracy of traditional machine learning and ensemble models by developing a novel, interactive consensus approach applicable to any supervised learning task.
- * Engineered a dynamic matrix for model convergence and implemented three distinct decision-making layers (soft consensus, hard consensus, and trust factor) to consistently outperform standard baselines.

- **Long Corpus Personality Detection** [📄]:

- * Proposed a novel approach to classify long documents of arbitrary length by leveraging two consecutive BERT models (Hierarchical BERT) to address transformer context limitations.
- * Successfully resolved GPU memory bottlenecks while fine-tuning the full pipeline of weights.

Data Scientist

Jun. 2022 – Jan. 2023

Dadmataech

Tehran, Iran

- Engineered automated ETL pipelines for Twitter, combining web crawling with regex-based text structuring.
- Developed a custom Django application for user data collection and management.
- Built and optimized a comprehensive suite of machine learning solutions, progressing from classical algorithms to advanced neural networks, transfer learning, and GPT-based language modeling.

PUBLICATIONS

Mehditabar, M., Rajput, S., & Sharma, T. (2026). "Watts This Smell: A Comprehensive Taxonomy of Software Energy Smells." IEEE International Conference on Software Maintenance and Evolution (ICSME, CORE A), IEEE, 2026. *Early acceptance rate: 6%*. [Paper]

Mehditabar, M., Rajput, S., Mastropaolo, A., & Sharma, T. (2026). "BRACE: Unified Benchmarking of Accuracy and Energy for Code Language Models." Journal Ahead Workshop at ICSE 2026. [Paper]

Karamdel, H., Ashtiani, M., Mehditabar, M. J., & Bakhshi, F. (2024). "A consensus-based approach to improve the accuracy of machine learning models." Evolutionary Intelligence, 1–22. [Paper]

Fatehi, S., Anvarian, Z., Madani, Y., Mehditabar, M., & Eetemadi, S. "MBTI Personality Prediction Approach on Persian Twitter." WiNLP at EMNLP 2022 Workshop. [Paper] [Poster]

HONORS & AWARDS

2025 – Winner, Lab2Market Industrial Proposal (\$10,000)

2025 – Research Scholarship Grant of \$43,000 for M.Sc. Program

2023 – Ranked 3rd out of 90 students in B.Sc. program

2019 – Ranked in the top 0.4% of 144,000 participants in the National University Entrance Exam

EDUCATION

Dalhousie University <i>Master of Computer Science - GPA: 4.3/4.3</i>	Halifax, Nova Scotia, Dalhousie <i>Jan 2025 – Present</i>
Iran University of Science and Technology (IUST) <i>B.Sc. Computer Engineering - GPA: 3.94/4</i>	Tehran, Iran <i>Sep. 2019 – Sep. 2023</i>